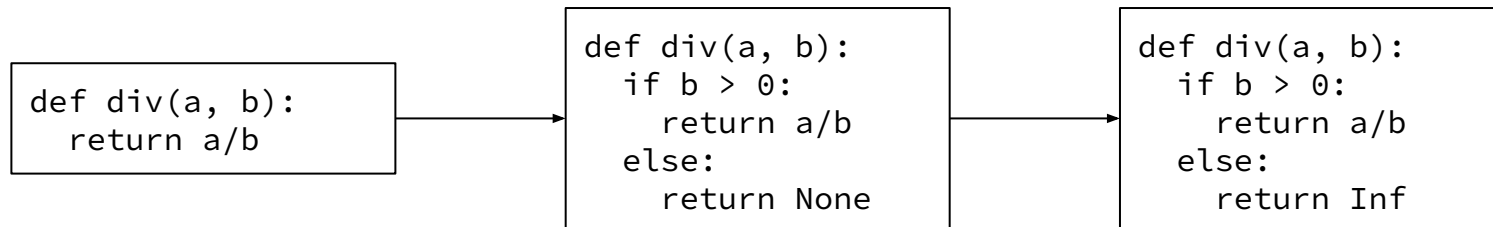


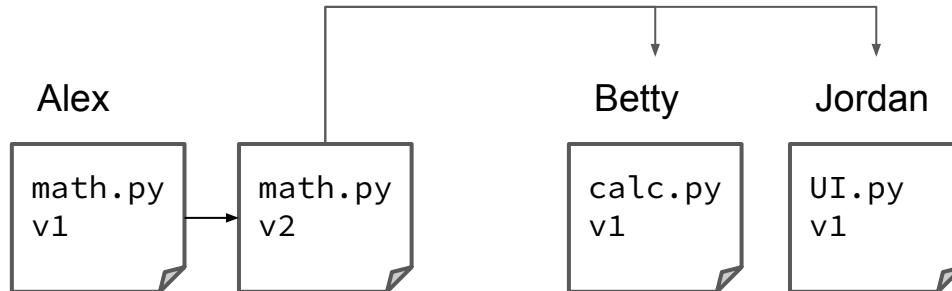
Version Control, Git, and GitHub

Software Engineering for Scientists

Version control manages changes to files for a project



- reverting back to an old version
- allowing developers to test changes without losing the original
- synchronizing code between developers and users



- tagging specific versions

Version control software



Software repository hosting company



git workflow

remote repository

local repository

local workspace

clone

commit

push

pull

git workflow

	remote repository	local repository	local workspace
clone			
commit			
push			
pull			

github

remote repository

lib.py

```
def div(a, b):  
    return a/b
```

github

remote repository

lib.py

```
def div(a, b):  
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clone



Alice

local repository

lib.py

```
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def div(a, b):  
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commit

clone

Bob

local repository

lib.py

```
def div(a, b):  
    return a/b
```

lib.py

```
def div(a, b):  
    return a/b  
  
def add(a, b):  
    return a + b
```

local workspace

lib.py

```
def div(a, b):  
    return a/b
```

```
def add(a, b):  
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commit

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remote repository

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commit

pull

merge

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lib.py commit history

github

remote repository

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setup.py

viz.py

run.py

repo commit history

github

remote repository

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setup.py

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\$ git clone repo grabs the most recent commit for every file

github

remote repository

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setup.py

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Recent commits
cause stability
issues

github

remote repository

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setup.py

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v1.0

Tag the commit state of
the repo to provide
consistency as
development continues

```
$ git clone repo --branch v1.0
```

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